Year 12 Maths - Motion with constant acceleration

Similar questions - Vertical motion under gravity

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A ball is thrown vertically upwards with speed 12 m s^{-1} from a point 1.5 m above horizontal ground. Take $g = 9.8$ m s^{-2} and ignore air resistance.

(a) Find the greatest height of the ball above the ground. **[3]**

(b) Find the time taken for the ball to reach the ground. **[4]**

(c) Find the speed of the ball as it hits the ground. **[2]**

(d) Find the length of time for which the ball is more than 5 m above the ground. **[4]**

2 Stone A is dropped from rest from the top of a vertical cliff. One second later, stone B is thrown vertically downwards from the same point with speed u m s^{-1}. Both stones hit the sea at the same instant, 3 seconds after A was dropped. Take $g = 9.8$ m s^{-2}.

(a) Find the height of the cliff above the sea. **[2]**

(b) Find the value of u. **[3]**

(c) Find the speed of each stone as it hits the sea. **[3]**

(d) State two modelling assumptions you have made. **[2]**

Question 1

(a) At the top $v = 0$: $0 = 144 - 2(9.8)s$, so $s = 7.35$ m above the throw point. Greatest height $= 1.5 + 7.35 = 8.85$ m (3 s.f.)

(b) Taking up as positive: $-1.5 = 12t - 4.9t^2$, so $4.9t^2 - 12t - 1.5 = 0$ and $t = \dfrac{12 + \sqrt{144 + 29.4}}{9.8} = 2.57$ s (reject the negative root)

(c) $v^2 = 12^2 + 2(9.8)(1.5) = 173.4$, so the speed is 13.2 m s^{-1}

(d) Above 5 m means $12t - 4.9t^2 > 3.5$. $4.9t^2 - 12t + 3.5 = 0$ gives $t = \dfrac{12 \pm \sqrt{75.4}}{9.8} = 0.339$ and 2.11. Duration $= 1.77$ s

Question 2

(a) $s = \frac{1}{2}(9.8)(3^2) = 44.1$ m

(b) B travels for 2 s: $44.1 = 2u + \frac{1}{2}(9.8)(2^2) = 2u + 19.6$, so $u = 12.25$ m s^{-1}

(c) A: $v = 9.8 \times 3 = 29.4$ m s^{-1}. B: $v = 12.25 + 9.8 \times 2 = 31.85$ m s^{-1}

(d) E.g. air resistance is negligible; the stones behave as particles; g is constant (and the cliff face is vertical, so neither hits it)